Potential Flow

- The concept of an ideal flow → inviscid fluid + incompressible + irrotational

The analysis of ideal flow: potential flow theory

$\vec{\omega} = \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k} = 0$ → the flow is irrotational

$$\omega = \frac{1}{2}\begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{bmatrix} = \frac{1}{2}(\operatorname{curl}\vec{V}) = \frac{1}{2}\operatorname{rot}\vec{V}$$

$$\omega_x = \frac{1}{2}\left[\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right] \qquad \omega_y = \frac{1}{2}\left[\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right] \qquad \omega_z = \frac{1}{2}\left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right]$$

$\operatorname{rot}(\operatorname{grad}\phi) = 0$ (i.e. $\operatorname{rot}\vec{V} = 0$)

$\vec{V} = -\operatorname{grad}\phi = -\nabla\phi = -\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\phi = \vec{V}$

- Velocity Potential

$u = -\frac{\partial \phi}{\partial x}$, $v = -\frac{\partial \phi}{\partial y}$, $w = \frac{\partial \phi}{\partial z}$

for 2Δ flow continuity eq $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow$ (laplace eq.) $\boxed{\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0}$

- Stream Function

$\frac{dy}{dx} = \frac{v}{u} \rightarrow u\,dy - v\,dx = 0$ Streamline function

$\psi = \text{constant}$ $d\psi = 0$

$d\psi = \frac{\partial \psi}{\partial x}\cdot dx + \frac{\partial \psi}{\partial y}\cdot dy = 0$ rotation in z

$v = -\frac{\partial \psi}{\partial x}$, $u = \frac{\partial \psi}{\partial y}$; $\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0\right)$ $\boxed{\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0}$

Prob. 1 - Determine if these 2Δ flows are irrotational or not.. Continuity check

a- $u = Cx$ $v = -Cy$

Continuity → $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 = C - C = 0 \Rightarrow$ flow is continuous

$\omega_z = \frac{1}{2}\left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right] \Rightarrow \frac{\partial u}{\partial x} = 0,\ \frac{\partial u}{\partial y} = 0 \Rightarrow$ flow is irrotational

b- $u = Ax^2 - Bxy$ $v = -2Axy + \frac{1}{2}By^2$

$\frac{\partial u}{\partial x} = 2Ax - By$ $\frac{\partial v}{\partial y} = -2Ax + By \Rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow$ flow is continuous

$\frac{\partial v}{\partial x} = -2Ay$ $\frac{\partial u}{\partial y} = -Bx$ $\Rightarrow \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) \neq 0 \Rightarrow$ flow is rotational

Prob. 2 - Determine that if these stream functions are possible in irrotational flow field?

a- $\psi = Ax + By^2$

$u = -\frac{\partial \psi}{\partial y}$ $v = \frac{\partial \psi}{\partial x}$; $\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) = 0 \rightarrow \omega_z = \frac{1}{2}\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right) = 0$ irrotational

$\omega_z = \frac{1}{2}\left[\frac{\partial^2}{\partial x^2}(Ax + By^2) + \frac{\partial^2}{\partial y^2}(Ax + By^2)\right] = \frac{1}{2}(0 + 2B) \neq 0 \Rightarrow$ flow is rotational

b- $\psi = Ax^2y^2$

$\omega = \frac{1}{2}\left[\frac{\partial^2}{\partial x^2}(Ax^2y^2) + \frac{\partial^2}{\partial y^2}(Ax^2y^2)\right] = \frac{1}{2}(2Ay^2 + 2Ax^2) \neq 0 \Rightarrow$ flow is rotational

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Prob. 3 - The velocity potentials (ϕ) in two dimensional flow are given. Determine the stream line functions (ψ).

a- $\phi = x + y$

$u = -\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}$ (velocity p. = streamline f.)

$v = -\frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$; $\frac{\partial (x+y)}{\partial x} = \frac{-\partial \psi}{\partial y} \rightarrow \frac{\partial \psi}{\partial y} = -1 \rightarrow \psi = -y + f_1(x)$, $\frac{-\partial (x+y)}{\partial y} = \frac{-\partial \psi}{\partial x} \rightarrow \frac{\partial \psi}{\partial x} = 1 \Rightarrow 1 + \frac{f_1(x)}{\partial x} \Rightarrow f_1(x) = x + c \rightarrow \boxed{\psi = -y + x + c}$

b- $\phi = x^2 - y^2$

$u = -\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}$ $v = -\frac{\partial \phi}{\partial y} = \frac{-\partial \psi}{\partial x}$; $\frac{\partial (x^2 - y^2)}{\partial x} = -\frac{\partial \psi}{\partial y} \rightarrow \frac{\partial \psi}{\partial y} = -2x$, $\frac{-\partial (x^2 - y^2)}{\partial y} = \frac{-\partial \psi}{\partial x} \rightarrow \frac{\partial \psi}{\partial x} = -2y \Rightarrow \psi = -2xy + f_1(x)$

$-2y + \frac{\partial f_1(x)}{\partial x} = -2y \Rightarrow f_1(x) = c$ $\boxed{\psi = -2xy + c}$